MusicVis new visualisations and default vis enhancements (WavePattern 2, Spectrum 2)

Test Number	1				
Test name	Spectrum MusicVis enhancement				
Description	Validate that the enhanced Spectrum musicVis works properly				
Test Status	Complete				
Test states	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
1.1	**The spectrum is radial and the lines properly move according to the frequencies of the music**	Step 1: Select the Spectrum2 from the menu. Step 2: Verify that the origin exists in the center of the window. Step 3: Select a music track Step 4: Visually verify that the spectrum reacts to the music.	The spectrum is displayed and reacts to the music dynamically.	Spectrum reacts as expected	Pass
1.2	**The spectrum changes colours from green to blue to red depending on the position of the line output.**	Step 1: Select the Spectrum2 from the menu. Step 2: Select a music track Step 3: Visually verify that the colours of the spectrum lines are correct.	The spectrum output lines are blue on the right, green on the left and red when the music is intense.	Spectrum colours are correctly responsive according to the music.	Pass
Test Number	2				
Test name	Wavepattern MusicVis enhancement				
Description	Validate that the enhanced Wavepattern musicVis works properly				
Test Status	Complete				
Test states	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
2.1	**The wavepattern2 properly incoporates a circle in the center and the waves react properly according to the frequencies of the music**	Step 1: Select the Wavepattern2 from the menu. Step 2: Verify that the wave exists in the center of the window. Step 3: Select a music track Step 4: Visually verify that the waves react to the music.	The waves react to the music dynamically.	Wavepattern reacts as expected	Pass
2.2	**The wavepattern circles are rainbow colour, indicating that the formulae for the colour randomiser works.**	Step 1: Select the Wavepatten2 from the menu. Step 2: Select a music track Step 3: Visually verify that the colours of the Wavepattern are correct.	The wave output lines are the correct colour.	Wavepattern shapes are displayed and colours react as expected	Pass
Test Number	3				
Test name	NEW spectrogram MusicVis				
Description	Validate that the new Spectrogram musicVis works properly				
Test Status	Complete				
Test states	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
3.1	**The Spectrogram music visualisation appears when chosen from the menu and reacts to the music**	Step 1: Select the spectrogram from the menu. Step 2: Verify that the spectrogram is being drawn in the center of the window. Step 3: Select a music track Step 4: Visually verify that the spectrogram lines react to the music and stay displayed	The spectrogram reacts to the music dynamically and displays the waves properly	Spectrogram musicvis reacts as expected	Pass
3.2	**The Spectrogram lines display and keep going, wrapping when the page is scrolled**	Step 1: Select the Spectrogram from the menu. Step 2: Select a music track Step 3: Visually verify that the lines wrap and output correctly.	The spectrogram outputs lines and wraps, deleting old lines	The spectrogram outputs and wraps as expected	Pass

Test Number	4				
Test name	NEW ConvergingLines MusicVis				
Description	Validate that the new ConnvergingLines musicVis works properly				
Test Status	Complete				
Test states	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
4.1	**The ConvergingLines displays the lines and particle effect properly react to the music**	Step 1: Select the ConvergingLines from the menu. Step 2: Verify that the lines are displayed in the bottom of the window and converge to a center point. Step 3: Select a music track Step 4: Visually verify that the particles and lines react	The lines display with randomised colours and particle effects generate and disappear properly when music is played	ConvergingLines musicvis reacts as expected	Pass
4.2	**The ConvergingLines is consistent with the viewport width, adapting the lines according to the width of the display.**	Step 1: Select the ConvergingLines from the menu. Step 2: Select a music track(optional) Step 3: Visually verify that the lines do not change height when the window is resized and the particles are always generated from the center of the canvas at the converging point.	The Converging Lines are always tethered to the converging point and the particles always generate from that area	The spectrogram outputs and wraps as expected	Pass

Test Number	5				
Test name	NEW RainbowCube MusicVis				
Description	Validate that the new RainbowCube musicVis works properly				
Test Status	Complete				
Test states	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
5.1	**The RainbowCube displays and draws the 3D object, also properly reacting to the music**	Step 1: Select the RainbowCube from the menu. Step 2: Visually Verify that the RainbowCube exists in the center of the window Step 3: Select a music track Step 4: Visually verify that the Cube reacts to the music	The RainbowCube reacts to the music by scaling and rotating depending on music intensity	RainbowCube musicvis reacts as expected	Pass
5.2	**The RainbowCube is consistent with the viewport width, adapting to the width of the display.**	Step 1: Select the RainbowCube from the menu. Step 2: Select a music track(optional) Step 3: Visually verify that the cube stays in the center of the window when resized	The RainbowCube stays in the center of the window when the window is resized	The RainbowCube reacts as expected	Pass

Section 2: Additions to the main menu & testing the new soundbar addition

Test Id	1				
Test name	Soundbar testing				
Description	Validate that the soundbar and all GUI elements are working as expected (song selector, play/pause button, scrubber, volume rocker, fullscreen toggle)				
Test Status	Complete				
Test cases	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
1.1 Initalising	**The soundbar appears with all associated elements.**	1. Open the music app. 2. Visually verify that the soundbar appears with all necessary GUI elements.	The Soundbar appears at the bottom of the window with all buttons.	The soundbar appears with all elements as expected	Pass
1.2 Global Soundbar	**The soundbar appears on all Music Visualisations**	1. Select any music Visualisation. 2. Visually verify that the soundbar appears. 3. Use every function on the soundbar to ensure they interact properly with the visualisation. 4. Repeat steps for each visualisation.	The soundbar appears after a new musicVis is chosen	The soundbar appears with all elements as expected	Pass
1.3 Music track Selection	**The 'Choose track' menu on the soundbar allows the user to choose music tracks to play and the track automatically plays when chosen. The old track should end when a new one is chosen from the same menu**	1. Click on the drop menu 2. Choose a track from the list 3. Verify that the audio is loaded and the chosen music track is automatically played. 4. Choose a different track from the drop menu 5. Verify that the previous track stops playing and the next one is loaded and automatically played. 6. Repeat steps for all available tracks	The drop menu opens revealing the music track options, the chosen track automatically plays. When a new track is chosen the previous track stops playing and the new track plays automatically	The drop menu appears and allows the user to choose a music track, automatically playing it.	Pass
1.4 Pause/Play button	**The pause and play button perform the desired functions**	1. Ensure a music track is already chosen from the drop menu 2. Click on the button to pause playback when the track is playing. 3. Click on the button to resume playback. 4. Repeat steps for all tracks	The music tracks can be paused and played when pressing the button without affecting or restarting playback.	The pause and play button work as expected	Pass
1.4 Scrubber slider	**The scrubber is able to fast forward or rewind the music depending on the slider position.**	1. Ensure a music track is already chosen from the drop menu and is playing. 2. Advance the slider to the middle of the music track and note the timestamp appearing next to the slider, it should display that the track is at half playback length. 3. Verify that the track has advanced forward through audio output. 4. Repeat steps 2 & 3 when audio is paused to playing.	The music track advances when the scrubber slider is moved	The scrubber advances the music track, however the music automatically plays when the scrubber is moved.	Pass(minor defect)
1.5	**The volume rocker changes the volume of the music properly**	1. Ensure a music track is already chosen from the drop menu and is playing. 2. Move the volume rocker up and down to desired volume. 3. Verify that the track has changed volume through audio output. 4. Repeat steps 2 & 3 when audio is paused to playing.	The volume changes depending on the position of the rocker	The volume rocker works as expected	Pass
1.6	**The Fullscreen button toggles fullscreen**	1. Press the fullscreen button 2. Visually verify that the window toggles to fullscreen.	The Window is toggled to fullscreen	The Fullscreen button works as expected	Pass

Test Id	2				
Test name	Main menu				
Description	Validate that the newly styled menu and prompt text appear and work properly				
Test Status	Complete				
Test cases	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
2.1 Initalising	**The Menu text "Press spacebar to open menu" appears and disappears when the menu is displayed**	1. Open the music app. 2. Visually verify that the text appears 3. Open the menu by pressing the spacebar 4. Visually verify that the text disappears	The text appears when the menu is closed and is hidden when the menu is open	The text element works as expected	Pass
2.2	**The newly designed Menu appears and disappears when the spacebar is pressed.**	1. Open the music app. 2. Visually verify that the menu is not displayed. 3. Open the menu by pressing the spacebar 4. Visually verify that the menu appears and displays the music visualisation options.	The menu appears and hides properly when the spacebar is pressed, displaying all musicVis info properly	The menu displays as expected	Pass
2.3	**The Menu allows choosing oof different visualisations by clicking and pressing keys 1-9.**	1. Open the music app. 2. Open the menu by pressing the spacebar 3. Click on the desired visualisation 4. Visually verify that the displayed MusicVis corresponds to the chosen MusicVis 5. Repeat steps 3-4 for all visualisations 6. Press the corresponding number on the keyboard to load the desired visualisation 7. Visually verify that the displayed MusicVis corresponds to the chosen MusicVis up to number 9. 8. Repeat steps 6-7 for all musicVis options	The menu responds to clicks and keys pressed and displays the correct corresponding music Visualisation	The menu works as expected	Pass

Test Id	3				
Test name	GUI menu hiding and functionality				
Description	Validate that the GUI works properly and hides and shows as expected				
Test Status	Complete				
Test cases	**Scenario**	**Steps**	**Expected Output**	**Actual output**	**Pass /Fail**
3.1 Initalising	**The Gui appears and manipulates the BlockMidHighLow visualisation properly**	1. Open the music app. 2. Visually verify that the GUI appears 3. Select a music track from the soundbar 4. Move each slider on the gui and visually verify that the sliders manipulate the movement and randomness of the line and blocks 5. change the colour of the box and line 6. Visually verify that the colour of the box and lines change	The Gui appears and is able to manipulate and change colours and speed of the reactivity of the visualisation	The GUI appears and works as expected	Pass
3.1	**The Gui appears and hides when double clicked**	1. Open the music app. 2. Visually verify that the GUI appears. 3. Double click the top of the GUI 4. Visually verify that the selector sliders and colour selectors are hidden 5. Double click the GUI window again to expand it. 6. Visually verify that the window expands and the selectors work as expected 7. Repeat tests on all Music Visualisations	The GUI appears and hides upon double click on all visualisations, with the buttons working properly after opening	The GUI displays and works as expected	Pass